La Scuola d'Italia / Dr. Huson / IB Math
28 May 2026

First & Last name:

8.4 Classwork: Rational Functions Challenge

A graphing calculator is allowed. Answer on lined or graph paper; label each part clearly. Show enough algebra that a marker can follow your reasoning — a GDC answer alone is not sufficient unless the part says "write down".

1. [Maximum mark: 8] The function f is defined by $f(x) = \dfrac{3x-1}{x-2}, \quad x \in \mathbb{R},\ x \neq 2.$

   (a) Write down the equation of the vertical asymptote of the graph of f. [1]

   (b) Write down the equation of the horizontal asymptote of the graph of f. [1]

   (c) Find the x-intercept and the y-intercept of the graph of f. [2]

   (d) Find $f^{-1}(x)$. [3]

   (e) State the domain of f^{-1}. [1]

2. [Maximum mark: 10] Consider the function $g(x) = \dfrac{2-2x}{x+2}, \quad x \in \mathbb{R},\ x \neq -2.$

   (a) Show that $g^{-1}(x) = g(x)$. [3]

   (b) Using your GDC, sketch $y = g(x)$ for $-8 \leq x \leq 6$. State the equations of the two asymptotes and the coordinates of both axis intercepts. [4]

   A function of the form $h(x) = \dfrac{2-3x}{cx+d}$, with $c, d \neq 0$, also satisfies $h^{-1}(x) = h(x)$ for all x in its domain.

   (c) Write down, in terms of c and d, the equation of the vertical asymptote and the equation of the horizontal asymptote of $y = h(x)$. [2]

   (d) Find the value of d. [1]

3. [Maximum mark: 8] The graph of $y = \frac{1}{x}$ is transformed by the following sequence, in order:

- a vertical stretch with scale factor 2,
- a reflection in the x-axis,
- a translation 4 units to the right and 3 units up.

The image is the graph of $y = p(x)$.

(a) Write $p(x)$ in the form $p(x) = \dfrac{ax+b}{x-4}$, where $a, b \in \mathbb{Z}$. [3]

(b) State the equations of the two asymptotes of $y = p(x)$. [1]

(c) Find the x-intercept and the y-intercept of the graph of p. [2]

(d) Find $p^{-1}(x)$ and state its domain. [2]

4. [Maximum mark: 6] Let $f(x) = \dfrac{1}{x-1}$, $x \neq 1$, and $g(x) = 2x+3$, $x \in \mathbb{R}$.

(a) Find $(f \circ g)(x)$, giving your answer in the form $\dfrac{1}{a(x-h)}$. State its domain. [3]

(b) Hence describe $(f \circ g)$ as a sequence of transformations of the parent function $y = \dfrac{1}{x}$. [3]